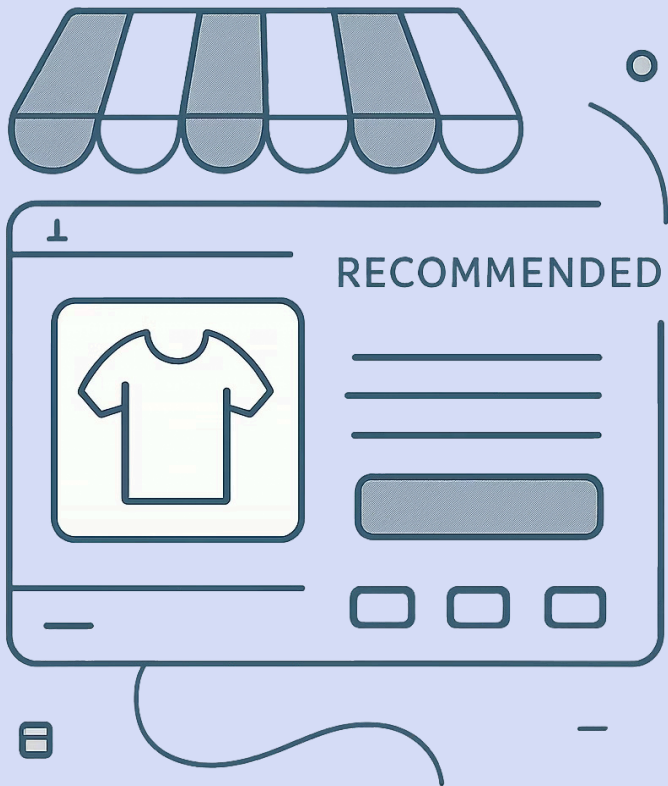




Multi-Modal Product Recommender

A data-product kickoff contract — replacing a popularity-based baseline with an intelligent, multi-modal recommendation system. Built collaboratively, measured honestly, and ready to ship.

Three Roles, One Shared Agreement

This requirement reflects what three personas agreed on — not one team's wish list.

Product Manager

Owns scope, sequencing, and a clear definition of "done" across the four lifecycle pillars.

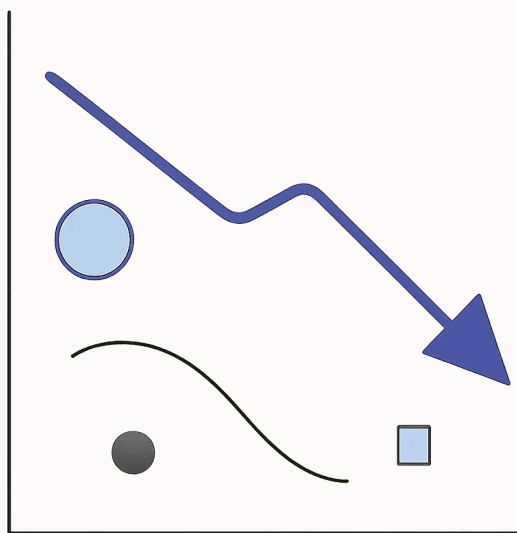
Commercial Stakeholder

Owns the business case. Wants revenue and engagement to move on the recommendation surface.

Senior AI Engineer

Owns model design and evaluation rigor. Demands defensible metrics, honest baselines, and reproducibility.

The Problem With the Baseline



What's in production today

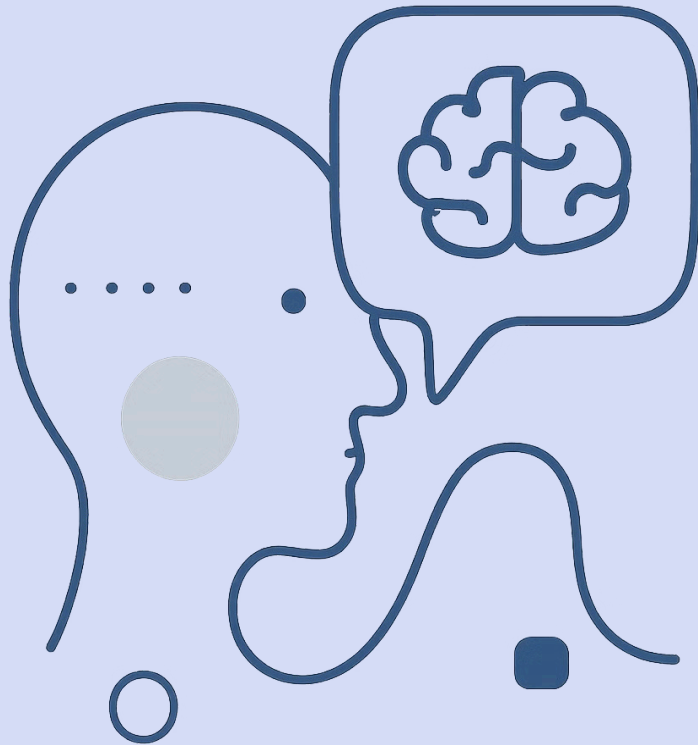
A **popularity-by-category recommender** — the most popular items within a category, served as suggestions. Trivial to ship. Zero training required.

Why it's not enough

Suggestions are **generic and category-obvious**. Revenue and engagement KPIs on the suggested-product surface are **flat**. The Commercial Stakeholder called for a replacement.

📄 **Honesty boundary:** Revenue and engagement are the motivation — not the measured outcome. This project uses offline metrics only, derived from a static product catalog.

How the New Recommender Works



1

User Query

Upload a photo **or** describe the item in natural language to an AI agent.

2

AI Agent Orchestrates

Chooses which tool to call: **classify_product** (vision model) or **search_similar_products** (retrieval index).

3

Two Signals Combine

Image classification filters and boosts the candidate set. **Metadata similarity** ranks by semantic relevance.

4

Ranked Recommendations

Final output: on-category, semantically relevant products — not just the most popular.

Results: The Acceptance Bar Is Met

Offline evaluation over 1,000 query items. The hybrid recommender decisively outperforms the popularity baseline on all three metrics at k=5 and k=10.

Metric	Baseline (Popularity)	Proposed (Hybrid)	Lift
Precision@5	0.325	0.898	+176%
Precision@10	0.308	0.873	+184%
NDCG@5	0.320	0.906	+183%
NDCG@10	0.309	0.888	+187%
Recall@5	0.003	0.022	+636%
Recall@10	0.006	0.037	+515%

❏ Recall is low by construction — proxy-relevant pools far exceed k. **Precision and NDCG are the meaningful metrics.** Relevance is content-based ground truth from catalog columns, not observed user behavior.



Prescriptive Recommendation: Ship It

1

Go Live Behind a Feature Flag

Deploy the hybrid recommender on the suggested-product surface with the popularity baseline as fallback.

2

Run an Online A/B Test

Validate the motivating KPIs — click-through rate and add-to-cart — before claiming business value.

3

Calibrate the Relevance Threshold

Define a recommendation cutoff that balances coverage with quality. Set OKRs and measure against them.